Detecting and Removing Gender Bias from Word Embeddings

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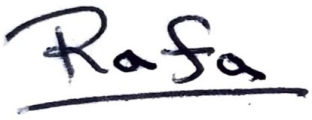
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Declaration

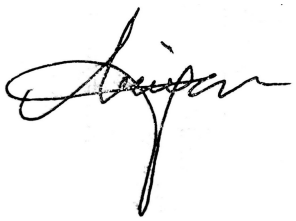
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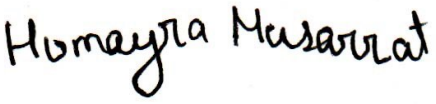
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Abstract

Machine learning has the potential to uncover data biases resulting from human error when it's implemented without proper restraint. However, this complexity arises from word embedding, which is a prominent framework for capturing textual input as vectors applied in different machine learning and processing natural language tasks. When this occurs, the dataset becomes skewed, and the ML model's objectives are compromised. This happens due to inaccurate results, biased assessments, and varying degrees of precision across contexts. To that intent, we plan to demonstrate a methodology for adjusting an embedding such that unintended associations, like the one between "receptionist" and "female," are eradicated while others, like "queen" and "female," are retained. Our techniques to "debias" an embedding are outlined, and we provide metrics to assess overall internal and external gender inequalities across all embeddings. To measure the extent to which implicit and explicit biases exist, we first identify a gender subspace and then evaluate the embeddings to see which of their components falls into that space. We intend to employ methods such as eliminating the embedding's component along the gender subspace to debias it, establishing a category of gender-specific combinations of words, and ensuring that each and every gender-neutral word is equally distinct from the gender-specific word pairs. These gender-neutral embeddings can then be used across several applications.

Keywords: Natural Language Programming; Bias; Debiasing; Word embeddings; Gender Bias

Table of Contents

Declaration **i**

Approval **ii**

Abstract **iii**

Table of Contents **1**

1 Introduction **2**

2 **3**

2.1 Research Statement 3
2.2 Research Problems 3
2.3 Research Objectives 4

3 Literature Review **5**

4 Work Plan **22**

5 Conclusion **24**

Bibliography **27**

Chapter 1

Introduction

Word embedding is considered a very important breakthrough in the field of NLP, as it has helped computers learn to interpret textual material at a deeper level. It is challenging to work with textual data given that computers or machine learning models lack human-level language comprehension.
Word embeddings, in their most basic form, are word representations that mediate between the human and computer perceptions of words. In the field of language analysis especially NLP, this term is used to describe how words are represented for text analysis. It is primarily used in the context of a real-valued vector. It has the ability to capture the meaning of the word to the point that it can anticipate that two words that are next to each other in a vector subspace are also semantically similar. Word embeddings have been demonstrated to capture important relationships between words by using the vector differences between them.
Word embeddings are crucial when embarking on a text categorization or sentiment analysis venture. In spite of this, it has been noted that word embeddings often reveal significant biases present in the initial data. In addition, recent research suggests that the results of activities that use neural word embeddings may be skewed toward one gender or the other. In particular, word embeddings often correlate masculine phrases with scientific keywords and feminine phrases with artistic keywords [9]. The closeness between word embeddings of gender-neutral sentences reflects the sexism in word-vector connections as well. Furthermore, some findings suggest that there is a frequent instance of bias in machine learning algorithms. According to Zhao et al. [8], word embeddings were discovered to magnify the inherent biases in the training data.
Word embeddings have been the subject of hundreds of academic studies, with widespread use. Conversely, neither of these studies has emphasized the severity of the sexism these embeddings have, which risks introducing biases of various sorts into operational systems. In this study, we try to fill in such shortcomings in current studies by creating a comprehensive yet efficient method for reducing sexism in word embeddings. Though there have been a number of attempts to combat the gender disparity present in word embedding, but so far none of them have been successful. For example, one method designed to examine word embeddings for sexism is the Word Embedding Association Test (WEAT) [7] . While biases have been demonstrated in several ways, it is still unclear how they are accumulated in large datasets. Which is why we make an effort to reduce the inherent bias that is found in many word embeddings.

Chapter 2

2.1 Research Statement

We will be using natural language processing (NLP) to structure algorithms that will detect bias from pre-trained word embeddings, and we will also be planning to mitigate that bias so that the output we get from the model will be accurate. This will be one of the outcomes of the research that we will be conducting and we would be dealing with both static and dynamic word embeddings.

2.2 Research Problems

The tendency of static word embeddings and word vector space models to mix words that might have several meanings into a single representation is one of the most significant problems. Neural word embeddings, like Word2Vec and GloVe, are biased, according to studies conducted by Yang & Feng [19]. Furthermore, when words are embedded in a way that favors one gender over another, it has a detrimental influence on algorithms that are farther downstream. Gender bias in word representations could have profound implications for society. This is a problem, for instance, for individuals who employ the embedding model to create gender-aware apps, such as a recommendation system for prospective employers to review applicants' portfolios. This is due to the fact that the automated system has the potential to discriminate against candidates on the basis of their gender, as indicated by their given name [12].
As was previously said, it is possible to assert that word embeddings that include biases might cause enormous difficulties in the processing of the output of a model. When a given model is trained with a corpus that has a bias against one of the two genders, the bias is transferred to the model itself. As a result, when the model processes data, it develops a bias toward one of the two genders. It is high time to deal with the issue of bias that is inherent in word embeddings, regardless of whether they are static or dynamic. This is because natural language processing (NLP) is becoming more in demand, and word embeddings are the core of NLP. As a consequence, during the course of our study, we want to reduce this bias in the word embeddings, which will ultimately lead to correct findings being created by a system or model.

2.3 Research Objectives

While researching the existing debiasing algorithms for word embeddings, we noticed that the majority of them were focused on static word embeddings. Works focusing on removing implicit bias from dynamic word embeddings were scarce. It is a lot more challenging to work with dynamic word embeddings. Biases in transformer models are also quite a new topic, and we wish to explore that as well. BERT (and other pretrained transformers) are used to essentially create a representation of their text input for further fine-tuning, and these representations are often superior to word embeddings. But as the good things get amplified here, so do the bad, and gender bias seems to flare up in these pre-trained models as well. Here are the above points in compact form for better understanding:

1. Analyze the currently existing debiasing techniques for static word embeddings (also explore bias in dynamic word embeddings)

2. Work on creating a debiasing algorithm for dynamic word embeddings.

3. Work on the bias that is introduced through transformer models which are used for text fine tuning.

4. Work with direct and indirect biases present in word embeddings.

5. To reflect the gender bias that is still prevalent in society. For instance w2vNEWS is an embedding which is trained on google news articles. In spite of being trained on one of the most accredited news sources w2vNEWS still displayed gender bias.

Chapter 3

Literature Review

Initially, we have collected some papers from trusted sources and reviewed them. The papers we gathered were from Google Scholars and our focus was on those papers that were highly cited indicating they are more reliable. We also mostly included papers that got published in renowned sources such as ACL Anthology, NeurIPS, MIT Press, etc. Since our topic is based on gender bias in word embeddings, we selected and reviewed only those papers or sections of a paper discussing gender bias in word embeddings.

To properly understand how bias originates in word embeddings and its full manifestation we have studied the paper by Brunet et al[9]. They created an efficient technique to assess the effect of eliminating a tiny portion of the training corpus on the bias from a given word embedding that has been developed by training on a dataset and a standard that can be used to measure bias. Instances of prejudice, including gender bias, in some of the most popular word embedding algorithms have been discovered, that may have a detrimental effect when used in machine learning systems. Although biases have already been shown using other methods, how biases are acquired from training data still needs to be determined. Two major subproblems were identified and tackled separately to determine the impact that removing a small portion of the training corpus would have on the resulting bias: evaluating the impact that perturbations to the training data have on the learned word embedding and gauging the impact that the word embedding has on the bias. [9] . First, they established the ideas of differential bias and bias gradient and formalized the issue of comprehending bias in word embeddings. Then, they demonstrated how word embeddings taught with the GloVe method may approximate the differential bias.

Brunet et al. [9] referred to B(w) as any bias measure that accepts an embedding of words and returns a numeric value. Let p represent one of several tiny subsets of the corpus that is considered. Let w' represent the word embeddings that were learned using the altered dataset $\tilde{C} = C \backslash p$. The differention of the bias inside component $p \subset C$ is defined as follows: $B(w) - B(\tilde{w})$. This represents component p's incremental contribution to the overall bias. When the bias metric B(w) and a word embedding w are both differentiable functions of the cooccurrence matrix X, the bias gradient: $\nabla_X B(w(X)$ can be considered as follows: $\nabla_w B(w) \nabla_X w(X)$. When just a small number of terms in the vocabulary affect the bias metric, the gradient tends to be sparse. Computing and research could then be done. If one

wants to know how much the bias might shift in response to a small change in X, one can use the bias gradient to get a linear approximation. This made it possible to use it for a rough approximation of the differential bias of document k.

A more efficient algorithm was developed by combining the differential bias equation with another equation that approximates the effect a certain corpus perturbation has on a specific word vector. This allows for an approximation of the differential bias of each text in the dataset without removing any entries from the corpora or retraining the embedding. To understand the result of their research we analyzed their outcome. They eliminated sets of documents to test their methods, causing bigger corpus changes. To evaluate the efficacy of their approach, they subtract each document set from the corpora and train a new embedding on this perturbed corpora, then compare the projected bias change to the actual bias shift measured. They also predicted and assessed using numerous embeddings with the same hyperparameters but different random seeds. They showed a selection of their experimental findings, mostly from NYT WEAT1 (science vs. arts). Also, however bias in WEAT2 (weapons vs. instruments bias) was not really present within their Wiki setup. To illustrate how little effect most documents were expected to have on the differential bias, they provided a histogram depicting the varied degrees of removal bias for each document in its setting at the NYT (WEAT1). To determine whether or not the biases introduced by the perturbations were significant, the default biases were calculated from the original embeddings, and a Welch's t-test was employed to evaluate the new values to them. Only two of the 36 sets of random perturbations were statistically different from the usual. They compared the WEAT effect magnitude in their NYT GloVe embeddings to the one discovered in the PPMI representation of the corpus and looked at the difference (count-based dispersed formulation including a positive point-wise mutual information matrix). It is easy to quickly calculate how the PPMI representation will shift in response to a co-occurrence perturbation (such as the removal of a document). Through this, they were able to search the whole corpus for the documents that influenced bias the most. They did, however, find that the papers discovered in this manner had far less of an effect on the bias than the ones discovered using our technique. The WEAT effect size in GloVe embeddings is obviously affected by the documents considered relevant by their technique. Next, they demonstrated how the same documents affect the bias in word2vec embeddings and other bias measures, finding that both the PPMI and the word2vec displayed distinct shift in magnitude as a result of the perturbations. It was observed that papers detailing the work of male engineers and scientists accounted for the vast majority of the documents with the highest levels of bias. They also found that the rarest terms in each of their WEAT subgroups were associated with astronomy or were in the publications that were the most biased. Interestingly, they found that the embedded representations of uncommon words were more vulnerable to changes in the corpus.

The paper by Bolukbasi et al.[5] first addressed the issue of gender disparity implicit in word embeddings and proposed a solution. The authors described a debiasing algorithm that would mitigate unintended gender biases in embeddings without altering their useful characteristics. This paper provided quantitative evidence indicating word embeddings have geometric biases that are reflective of firmly ingrained

gender preconceptions. Even though a lot of research has been done on word embeddings, none of it shows how blatantly sexist they are or how they could cause bias in real-world systems. Ads displayed to users of different sexes on a number of popular websites display clear signs of gender bias. That is why it was important to solve this problem. Now, to solve this problem, this paper proposed two methods: Hard and Soft debiasing. Firstly, they compared a vector representation to the vectors of two gender-specific words in order to measure any inherent bias. For example, if the vectors of a man and a woman are not at the same distance from the vector of a nurse, this shows bias. Using cosine similarity, we can figure out how far apart or how similar two vectors are. They learned a gender subspace in the embedding using gender-specific terms like she, he, father, mother, brother, and sister, and the debiasing algorithm will only apply to and remove biases from gender-neutral terms while preserving the meanings of gender-specific terms. Due to the relatively small number of gender-distinct terms, it was found to be more sensible to consider the collection of gender-specific words as 'S' as well as consider the set of gender-neutral phrases to be the complement, (N=W/S). Using these gender-specific words, they trained a linear SVM classifier to get all gender-specific words. Using all gender-specific terms, they searched the corpus for gender-neutral terms such as programmer, housewife, and gardener. The gendered and non-gendered words in the language were now clearly distinguished.

They then chose gender-specific paired words to serve as the x-axis. For demonstration, they chose the words pair she-he as the beginning and ending points of the x-axis and mapped all other words onto this gender axis. This step assisted in identifying the direction of embeddings that capture bias. Now came the two ways to get rid of bias in the embedding: Hard Debias (Neutralize and Equalize) or Soft Debias. Hard debiasing is a severe procedure in which all references to gender are removed from otherwise gender-neutral terms. Neutralize ensures that the gender subspace contains no gender-neutral words. They simply projected all of the gender-neutral terms onto the y axis, removing the gender factor. Equalize guarantees that any impartial phrase is equally distant from all other phrases in each equality set. This happens by perfectly equalizing sets of words beyond the subdomain from the gender specific terms(he-she) functioning as the axis. The main drawback of Equalize is that it eliminates differences that might be useful in some contexts. Soft Debias is an approach that minimizes discrepancies between these sets while preserving as much resemblance as possible to the initial embedding, with the degree of similarity being controlled by a parameter. Depending on the values of the hyperparameter, soft debias will translate the gendered terms into a neutral one.

To test the effectiveness of the solutions the authors carried out various experiments to deduce if the solution meets certain criterias. Analogy generation task was used to assess the effectiveness of their debiasing algorithm. They programmed a system to produce a set of phrases that are analogues to "she" and "he" and polled the crowd to see if the results confirmed existing gender stereotypes. 19% of the top 150 analogies on the initial w2vNEWS embedding were thought to exhibit gender stereotypes by the majority of the 10 workers that evaluated the word embeddings. Following the use of our hard debiasing approach, just 6% of the revised word embedding were determined to be stereotypical.

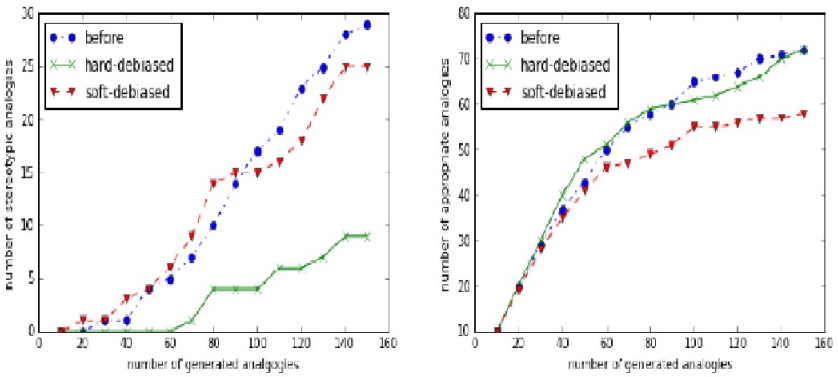


Figure 3.1: Left- stereotypical analogies and Right- appropriate analogies

Let us use the analogy in which he is towards the doctor as she is towards X to illustrate. As opposed to the previous embedding, which yielded X = nurse, the hard-debiased embedding now finds X = doctor. Gender-specific parallels were kept using the hard debiasing algorithm; for example, a woman with ovarian cancer would be similar to a man with prostate cancer. This proved that the embeddings' quality is maintained. Soft-debiasing, on the other hand, was not as successful as Hard-debiasing in eradicating gender bias.

However, two flaws in Balukbasi et al.'s [5] research were identified by Zhao et al. [12]. To begin with, the technique is simply a pipeline strategy that calls for a classifier to first identify gender-neutral phrases before resorting to the projection. The model's efficiency will suffer if the classifier makes an error and that mistake is then passed down. Second, their approach eliminates any traces of gender from terms that are vital in certain fields [12]. To address these shortcomings, they presented GN-GloVe, a gender-neutral variant of GloVe, for training word embedding models that include sensitive information such as gender. They focused on obtaining gender information without compromising the embedding model in one of their works. The aim was to keep gender information in selected word vector dimensions, which can help model mood and gender beyond binary. The motivation behind their work was that human-generated text embeddings can misrepresent gender, which substantially impacts downstream implementations. Biased embeddings may impact daily services. Developers using the embedding framework to build gender-sensitive apps like portfolio sorting or employment recommendation systems may worry about the automated system's bias against candidates. They proposed a method to get a solution to this problem they faced. They used GloVe as its foundational embedding model, preserving gender as its key variable [12]. Their method is adaptable and can be employed with a wide variety of embedding models and characteristics. The quality of the word embeddings were also calibrated or tested by the benchmark tests. Word similarity tasks compare word embedding models to human-annotated rating scales for word similarities. They used the same methodology as GloVe to build a word-to-word co-occurrence matrix, which they referred to as X. In this matrix, j

represents the number of occurrences of the i-th word in context j. Their embedding model divides a word vector into two parts: $(w^{(a)})$for neutralized components and $(w^{(g)})$ for gendered components. They then suggested reserving the gender characteristic, often called a "protected attribute," in the variable $(w^{(g)})$ as a means of achieving gender equality. As a result, the gender does not affect the meaning of the information stored in $(w^{(a)})$. They categorized all of the terminology into one of three categories, according to the gendered definitions that they found in WordNet: male-definition Ω_M, female-definition Ω_F, and gender-neutral Ω_N [12]. Following the aforementioned considerations, they developed their minimization goal. It is composed of three components: the first portion encapsulates the idea of proximity in a single word, while the other two sections attempt to restrict the gender information included in $(w^{(g)})$ in a way that does not influence $(w^{(a)})$. They examined two distant metrics and developed two distinct kinds of objective functions by making use of seed words for male and female definitions, respectively denoted by the symbols Ω_M and Ω_F. While one of the functions directly minimizes the negative distances between the words in the two groups, the other function caps the values of a word vector at 1 and -1, forcing $(w^{(g)})$ to take an extreme value. The final phrase advocates for words in Ω_N to keep their $(w^{(a)})$ in the null space of the gender direction v_g, where v_g is calculated by averaging the disparities between female and male words in a predetermined collection.

For experimentation, they have compared GN-GloVe to GloVe and Hard-GloVe. GN-GloVe separates the gender trait that is being safeguarded from the other elements that are inactive. On a newly annotated dataset, they test GN-GloVe's ability to make a distinction between words that define gender and words that fit a stereotype. GN-GloVe does a good job of figuring out how close two words are on typical benchmark datasets. GN-Glove eliminates gender bias in coreference resolution. They referred to the process of removing gender bias from GloVe as "Hard-GloVe," and it was done using the post processing approach. Every single one of their embeddings has been trained using the default hyper-parameters from the English Wikipedia dump from 2017.

In conclusion, they demonstrated that GN-GloVe maintains gender links in $(w^{(g)})$, and they illustrated it by utilizing $(w^{(g)})$ as the x-axis and a random value with words for the y-axis.The projection for GloVe is 0.080, the projection for HardGloVe is 0.019, and the projection for Gn-Glove is 0.052. GN-GloVe has been shown to reduce biases by 35%. Hard-GloVe is not as adept as GN-GloVe when it comes to differentiating between gender-stereotype phrases and gender-definition terms. When comparing gender-defining word pairs to "he - she," GN-GloVe finds a 97.7% accuracy rate. On the other hand, GloVe and Hard-GloVe are much more prone to errors. To further demonstrate its ability to generalize the gender pairs on the training set to identify other gender-definition word pairs, GN-GloVe obtains much better performance than Hard-Glove and GloVe on the subset. Their models are evaluated using the OntoNotes 5.0 and WinoBias datasets. GN-GloVe achieves results that are comparable to those of Glove and Hard-GloVe on OntoNotes while also reducing bias on WinoBias. GN-GloVe $(w^{(a)})$ minimizes coreference bias when only $(w^{(a)})$ is used to represent words. GNGloVe performs better in similarity challenges and maintains word proximity in analogy scores.

Next, we studied the paper authored by Dev and Phillips [14] which improves the solution provided in the paper [5]. However, they are capable of magnifying and conveying sexism, that could result in discrimination in a number of applications. According to research, word embeddings are likely to reflect the bias present in the data they are retrieved from. Furthermore, the work by Zhao et al. [8] in the paper showed that the output of machine learning algorithms exhibits more bias than the data from which it was created. Word vector embeddings are employed in machine learning for tasks that have a substantial impact on people's lives, including credit evaluation, crime prediction and other new fields, including evaluating loan applications, resumes for jobs, and college applications. Therefore, it was crucial that efforts be made to recognize and, if feasible, eliminate the bias that is inherent in them. To remove the bias from words that are overtly biased toward one gender over another, the study demonstrated easy methods for doing so.

In order to reduce bias a viable and widely applicable alternative is to first gather all word-vectors based on popular name and then take their linear projections This study simplifies, quantifies, and fine-tunes these approaches. Furthermore, the paper also showed how all word vectors can be projected simply linearly along a bias direction. In particular, it showed that these results may be slightly enhanced by minimizing the projection of words-vectors that are very distant from the projection distance. The Hard Debiasing method of Bolukbasi et al. [5] is somewhat more advanced and partly depends on community sourcing, whereas basic linear projection is more elective. The paper also shows two simple methods as an alternative to the hard debiasing method. (1) Subtraction : As a basic starting point, deduct the gender direction v_B from all word vectors $w' = w - v_B$. (2) Linear Projection : It has a different starting point which is more sophisticated than the previous. The starting point is to orthogonally project each word $w \in W$ onto the biased vector v_B.

To determine whether the offered procedures can genuinely mitigate bias from the data, various Quantifying tests are run on the data. The tests were the WEAT, the embedding quality test, and the embedding coherence test. It was demonstrated that the dampened technique performed better on the Google analogy test even when ECT, EQT, and WEAT scores seemed to be in identical ranges. The final results demonstrated that the user can debias data using any of the suggested methods while maintaining as much structure as is feasible, but that linear projection performed about as well as these dampening methods.

Next we read the paper by Wang et al. [23] which introduced us to Double-Hard Debias. Double hard Debias builds on Hard Debias. This reduced frequency's effect on gender direction. According to the tests conducted by authors, Double Hard Debias reduced gender bias without compromising word embedding quality. W was the word embedding lexicon to be debiased. Each word in W had a word embedding vector $\vec{w} \in R^n$. $B = \{b_1, \ldots, b_k\} \in R^n$ was generated by k orthogonal unit vectors. Vector v was projected on B using, $v_B = \sum_{j=1}^{k} (v \cdot b_j)\, b_j$

Bolukbasi et al. [5], assumed a collection of gender-neutral words ($N \subset W$). They also supposed a predefined array of n male-female word pairings ($D_1, D_2, \ldots, D_n \subset W$). Hard Debias initially finds a gender bias subspace. Here, $\mu_i := \sum_{w \in D_i} \vec{w} / |D_i|$. SVD(C)'s first k rows are the bias subspace B. Here,

$\mathbf{C} := \sum_{i=1}^{m} \sum_{w \in D_i} (\vec{w} - \mu_i)^T (\vec{w} - \mu_i) / |D_i|$. Bolukbasi et al. [5] set k = 1, which they also followed. Thus, subspace B is gender-directed. Hard Debias neutralizes word embeddings by changing each $(\vec{\omega})$ to zero projection throughout every word's gender subset $(\omega \in N)$. They have re-embed $(\vec{\omega})$ Within each phrase in $(\omega \in N)$: $(\vec{w} := \vec{w} - \vec{w}_B)$ They used the Neighborhood Metric in [23] which quantifies bias without gender direction by comparing words. They chose k out of the most distorted male and female words based on their embedding cosine similarity and gender direction. (W_m) and (W_f) are gender-biased terms. They assigned $(g_i = 0)$ if $(\omega_i \in W_m)$. For $(\omega_i \in W_f)$, $g_i = 1$. They clustered the embeddings of specific keywords using KMeans (k = 2) and they calculate gender labels as input, they determine a synchronization score of a which is: $a = \frac{1}{2k} \sum_{i=1}^{2k} 1 [\hat{g}_i == g_i]$

Here, they assumed a = max(a, 1 − a). When a= 0.5 means totally unbiased word embeddings. While the value of a is significantly closer to 1, it indicates a rise in bias. They used Double-Hard Debiasing where they selected 500 male and female biased words from the initial GloVe embeddings. Later they applied PCA across all of the word embeddings and picked the most relevant parts to discard. They projected embeddings into orthogonal spaces for each possible direction of (u). Hard Debias debiases embeddings in this intermediate subspace. They clustered these words' debiased embeddings and calculated gender alignment accuracy. This showed if projecting (u) away improved debiasing. Clustering efficiency for Wikipedia-trained GloVe embeddings dropped significantly when the projection anywhere along second core element was removed.

They experimented with the results of their method using datasets. Following Pennington et al. [2], 300-dimensional GloVe embeddings pre-trained on the 2017 January English Wikipedia dump of 322,636 unique words were used there. To put their theory to the test, they employed a wide variety of variants of the GloVe, GN-GloVe, GN-GloVe(wa), GP-GloVe, GP-GN-GloVe, Hard-GloVe, Strong-GloVe, and Double-Hard GloVe. Now to evaluate the performance of Double Hard Debias, the approach was effective for both diverse applications as well as encoding analyses. They implemented debiasing in downstream applications using coreference resolution. On OntoNotes 5.0, they trained a model for complete-sentence coreference resolution [23] with unique word embeddings, and then tested it on WinoBias. Less biased coreference systems had lower Diff values. Double-Hard GloVe had the lowest Diff in WinoBias. Double-Hard GloVe performed similarly to GloVe on OntoNotes, demonstrating that their technique preserves word embeddings. Double-Hard GloVe reduces gender bias and improved type-2 phrase performance from 75.1% to 85.0%.

During debiasing somewhere at embed level, the Word Embeddings Association Test was utilized (WEAT). Here, effect sizes (d) and p-values were estimated (p). A p-value greater than 0.05 indicated the absence of bias. P-values showed bias significance. A high p-value (greater than 0.05) implied no bias. Double-Hard GloVe routinely beat debiased embeddings. For career, family and science and arts, Double-Hard GloVe produced the smallest effect size and the smallest amount of bias. In the WEAT test, GN-GloVe amplified gender bias for Math & Arts words, but original GloVe embeddings did not. Double-Hard GloVe was unaffected. Hard Debias reduced bias more than other baselines. The fact that Double-Hard GloVe grouped the top 100, 500, or 1000 biased terms with the lowest precision demonstrates that the proposed strategy decreases gender bias. Double-Hard GloVe merged male and female word embedding the most after debiasing, hence disclos-

ing the least amount of gender information. They did an analysis of retaining word semantics. Double-Hard GloVe performed well and marginally outperformed other debiased embeddingss and could preserve word closeness. For concept categorization, Double-Hard GloVe matched GloVe embeddings as it retained semantic information in word embeddings.

Chaloner & Maldonado [13] tried to quantify gender bias across word embeddings throughout domains and identify new misleading subcategories for words. They believed that it was the first time that a non-monitored strategy for finding new subcategories of gender bias phrases by using word embeddings had been found. The reason behind their work was that all neural NLP systems mostly use word embeddings as input. These embeddings capture human biases, such as gender bias, which may affect the decisions these systems make.

There appears to be little research on how different types of gender bias manifest in word embeddings across domains. They applied the WEAT test for gender bias detection to four sets of word embeddings taught in libraries spanning four categories: social media (Twitter), a Wikipedia-based gender-balanced corpus (GAP) and a biomedical corpus (PubMed) and news (Google News). They used five categories of words - i) Career vs. Family activities, ii) Maths vs. Arts, iii) Science vs. Arts, iv) Disparities in personal descriptions based on intelligence vs. appearance, and v) Physical or emotional strength vs. weakness to test for the presence of gender bias in the aforementioned domain corpora.

The input for WEAT Hypothesis testing is a presumed gender discrimination keyword classification through 2 groups of target words, X = programmer, engineer, scientist and Y = nurse, teacher, librarian. The null hypothesis was examined by Chaloner & Maldonado [13] to determine if X or Y is far more biased toward one group or if there exists no difference overall bias between the two lists. Two word reference lists, M = man, male, he and F = woman, female, she, are used to compare bias. These words, also known as attribute words, clearly reflect male and female conceptions.

A permutation test is used to test the hypothesis H_o. $(X \cup Y)$ is thoroughly divided into alternate objective lists $\hat{X}$ and $\hat{Y}$, and the partial p-value, or the proportion of permutations where the analysis factor occurs s($\hat{X}$, $\hat{Y}$, M, F) surpasses the measured analysis factor s(X, Y, M, F) is computed. Such a value for p represents the likelihood that H_o is true. In other words, there is a chance that there is not any distinction between X and Y (in regard to M or F) so that the word class is consequently unaffected by bias. The bias decreases with increasing p-value. If the p-value for a word category is less than the threshold of 0.05, it is considered to possess statistically significant gender bias.

Additionally, Chaloner & Maldonado [13] proposed a method for automatically providing unprecedented subcategories of gender-biased words within an embedding set of words. They used K-Means++ clustering, which was fast and produced clusters of comparable sizes. The n highest male- and female-associated keywords were collected for each cluster. These are the gender bias word categories that have been found. To make the computation manageable, they utilized the entire WEAT hypothesis testing procedure, employing randomized experiments with 1,000 repetitions per grouping to assess if the candidates' bias is statistically significant.

After applying WEAT Hypothesis on the four domains - Google News, Twitter, Pubmed and GAP based on the five categories of words mentioned previously, they gathered few outcomes. From Google News, gender bias in all five categories was significantly detected. From Twitter, bias was only detected in the career vs family category. The majority of effect sizes (Cohen's d) were less than one, showing substantially weaker connections with gender-specific attribute terms. Bias was seen the least from PubMed, but gender bias exists in biomedical studies. From GAP, which is constructed off of a gender-biased original text (Wikipedia), gender bias was found at a lower rate than anticipated. This may be due to the fact that GAP's vocabulary excludes a significant number of the characteristic and objective word lists utilized on assessments. For each cluster, the returned gender bias word categories candidates seemed somewhat cohesive. It was able to identify conceptually consistent clusters of gender-related terms. Nevertheless, many if not most terms seemed to be gender biased in a negative sense. Upon applying the WEAT hypothesis testing technique to the candidates, overall clusters throughout all embedding types gave a p-value less than 0.001, indicating strong biases.

Garg et al. [10] investigated how the geometric structure of word embeddings changed over time, concerning racial stereotypes and gender, in the United States. A research that was conducted recently suggests that word embeddings, in addition to a wide variety of other machine learning approaches, capture popular preconceptions. This is due to the fact that these sorts of preconceptions may be found across the enormous library of training texts. If the embedding method is utilized for sensitive applications like search rankings, product recommendations, or translations, this might be problematic since the embedding algorithm automatically learns these prejudices.

To solve this problem, 100 years of data training worth word embeddings were analyzed. They built a metrics-based framework on word embeddings in order to determine how gender stereotypes and behaviors in the United States towards women of ethnic minorities altered during the 20th and 21st century. In their investigation, the researchers employed the typical Google News word2vec vectors that had been trained on the Google News dataset [10]. They employed the Google Corpus of Historical American English (COHA), which had been trained before, in order to conduct a historical study utilizing the COHA and Google Books [10] . In addition to that, the GLoVe method [2] and embedding from the New York Times Annotated Corpus [1] were used by the researchers. The professions category makes use of historical data from the United States Census [3].

By using word embedding and word lists, one is able to quantify the degree of connection, or embedding bias, between words that do not belong to the same group and between groups of words. The data, when analyzed, revealed a substantial association between the ingrained gender and ethnic occupational prejudices, on the one hand, and the actual profession participation rates, on the other. During the same time, they demonstrated how adjective relations in the embedding explain how different groups of persons are seen throughout the course of time. In addition, the outcomes of their research indicated that, with the assistance of word embedding, it is possible to systematically quantify popular prejudices as well as other historical trends.

Kumar et al. [22] found that according to recent studies, word embeddings exhibit gender, racial, and religious biases. Therefore, in their research they plan to mitigate gender bias from pre-trained word embeddings. It was seen that words like "nurse" are more linked to women and words like "doctor" are more linked to men. Furthermore similar results were seen from a model consisting of word embeddings which was used to train a known social media network [22] .

To solve the problem of bias in word embeddings, [22] proposed using RAN-Debias. This technique is a well known and effective approach to debias non contextualized word embeddings. They used this to address a variety of methods including repulsion and attraction; along with this debiasing based on neutralization was also pointed out. In addition to reducing the projection of gender-biased word vectors in the gender direction, RAN-Debias minimized the semantic similarity between adjacent word vectors that had unlawful proximities, which decreased the semantic similarity with nearby word vectors that have illegal proximities. Regarding the selection of the group of words that need to be debiased, they also suggested KBC (Knowledge Based Classifier), a word classification technique. In order to achieve more accurate classifications, KBC drew on a number of lexical knowledge bases that were previously in existence. In addition, they offered the Gender-based Illicit Proximity Estimate (GIPE), a metric that measures gender bias in the embedding space as a result of illicit proximities between word vectors.

The results showed that the methods were indeed helpful in mitigating bias from pre-trained word embeddings. They assessed the effectiveness of debiasing using different evaluation indicators. RAN-GloVe which is basically GloVe word embeddings on which RAN-Debias method was applied, exceeded the previous baseline standard, which was GN-GloVe by 21.4% tested on the SemBias dataset, in the gender stereotype type for the gender relational analogies test [12]. According to trials based on a range of assessment measures, RAN-Debias significantly outperformed the latest one in reducing proximity bias (GIPE) by at least 42.02%. This was a significant achievement. In addition, the performance of RAN-GloVe on word analogies and similarity tasks across a variety of benchmark datasets demonstrated a modest disturbance to the semantic flow.

Kaneko & Bollegala [17] proposed a debiasing method that firstly preserves gender-related information in feminine and masculine words. Furthermore, they maintain the impartiality of gender-neutral phrases and minimize the bias of stereotyped words. Therefore, they thought an accurate and efficient debiasing approach that can successfully debias a pre-trained word embedding was essential.

Consequently, they proposed a debiasing method that is able to differentiate between stereotypical biases from the desirable bias encoded with the gender information in words. They utilized a set of pre-trained d-dimensional word embeddings as $\{\boldsymbol{w}_i\}_{i=1}^{|\mathcal{V}|}$ over a vocabulary $\mathcal{V}$ and projected that embedding into a debiased l-dimensional space described as $(E : R^d \rightarrow R^l)$. They used E(w) to describe the new vector representation. The Vocabulary $\mathcal{V}$ is divided into four mutually exclusive sets where $\omega_m \in \mathcal{V}_m$ (Masculine), to preserve its masculine properties, $\omega_f \in \mathcal{V}_f$ (Feminine), to preserve its masculine properties, $\omega_n \in \mathcal{V}_n$ (Neutral), to preserve its

gender neutrality; and lastly, $\omega_s \in \mathcal{V}_s$ (Stereotypical), to remove any stereotypical bias. A denoising autoencoder captures the pre-trained embedding, and a decoder reconstructs word embeddings similar to the original using the encoder's debiased embedding eliminating gender biases.

Their overall model was strictly trained so that it could satisfy the aforementioned constraints. They defined a Feminine Regressor as $C_f : R^l \rightarrow [0, 1]$, parameterised by (θ_f) and a Masculine Regressor as $C_m : R^l \rightarrow [0, 1]$, parameterised by (θ_f). Both of these Neural Networks predicted the degree of masculinity or femininity of a word w. Highly feminine/Masculine words had values closer to 1. $(E : R^d \rightarrow R^l)$, parameterised by (θ_e) is the encoder, which also acts as the debiasing function that eliminates gender stereotypes from embeddings. $(D : R^l \rightarrow R^d)$ parameterised by (θ_d) is the decoder that translates the l-dimensional space into the original embedding but without gender bias.(C_f) and (C_m) are both implemented as feed-forward neural networks with one hidden layer and use the sigmoid function as the nonlinear activation function. Similarly, Encoder E and Decoder D are feed forward neural networks with two hidden layers and hyperbolic tangent is used as the activation function. The parameters of the neural networks E, D, (C_f), (C_m) were trained by minimizing the loss function. The encoded space needs to retain the gender information with regard to the Feminine and Masculine words. If their losses are denoted by L_f and L_m then their squared losses indicate how much the constraints were satisfied.

Their model required embedding the stereotypical and gender-neutral words orthogonal to the gender direction vector (v_g). The value v_g is computed using a set of feminine and masculine word pairs denoted by (Ω). Loss, L_g is computed using the squared inner product between v_g and the debiased stereotypical or gender-neutral words. To protect the semantic meaning encoded within the original word embedding they minimized the loss during the reconstruction of the word embedding. L_r is denoted as the reconstruction loss. The total objective, $L = \lambda_f L_f + \lambda_m L_m + \lambda_g L_g + \lambda_r L_r$, where, $\lambda_f, \lambda_m, \lambda_g, \lambda_r$ are all hyper-parameters that determine the relative importance of different constraints. Similar to Kumar et al. [22] SemBias dataset created by Zhao et al. [12] is used to evaluate Kaneko & Bollegala's [17] proposed debiasing method. Sembias instances had 1 gender-definition word pair, 1 gender stereotype pair, 1 gender definition pair, and 2 similar-meaning word pairings unrelated to gender. Their method worked better on definition and stereotype word pairings. Which is why Kaneko & Bollegala's [17] proposed method works very well in conjunction with GN-GloVe to remove any residual bias as it performs better in Gender Neutral GloVe than in GloVe.

To evaluate the semantic accuracy they performed two tasks, such as Semantic Similarity Measurement and Analogy Detection. According to Kaneko & Bollegala's [17], Semantic and Syntactic analogies in google analogy datasets , MSR dataset and SemEval dataset are used to benchmark the analogy detection capability of the reconstructed debiased embedding. No significant degradation of performance is seen due to the debiasing of the proposed method. Current standard benchmark datasets used to detect semantic similarity contain biased examples because these were not created keeping gender stereotypes in mind. To solve these they created a new dataset by balancing the original dataset by adding extra word pairs generated from the opposite sex with similar human ratings. After the test, they assessed that the performance of their proposed solution(GP(GloVe)) is similar to GloVe which

proves that bias is removed while keeping the information in the embedding intact.

Yang & Feng's [19] came up with one of the latest post-processing methods to remove both gender biases. Neural word embeddings can be biased, and tasks that use them may give gender biased results. Their approach also enhanced word vectors and lowered gender bias in subsequent concerns. This method excels in gender bias reduction, word and phrase evaluation, and coreference resolution. Their solution also addresses past restrictions.

The word vectors of gender-neutral words tend to favor the spatial directions of either "male" or "female" words, known as "gender bias associated with gender direction." Distance between gender-neutral word embeddings causes gender bias. Yang & Feng's [19] method is light and straightforward to use. They have used the statistical relationship between gender definition word embeddings and gender-biased word embeddings to make a simple and causal way to reduce gender bias in word vector relations. Gender-debiasing is achieved by subtracting the learned, misleading gender data from gender-biased word vectors. Learning gender information in gender-biased word vectors approximates and eliminates word vector relation and gender direction bias. They extracted gender-neutral semantic information from word vectors using non-gender-defined word vectors. They proposed that the debiased non-gender-definition word vectors $\hat{V}_N$ are learned by eliminating the approximate gender information $\hat{G}$ from the initial $\hat{V}_N$. So, they got the value $(\hat{V}_N := V_N - \hat{G})$.

Here $\hat{G}$ is obtained by estimating $(\hat{V}_N)$ using gender-definition word vectors $\hat{V}_D$, so they got $\hat{\mathbf{G}} := E\left[\mathbf{V}_N \mid \mathbf{V}_D\right]$. $\hat{G}$ learns gender via $\hat{V}_D$ predictions. $\hat{G}$ doesn't learn the semantic information of $\hat{V}_N$ when approximating it with $\hat{V}_D$. $\hat{G}$ only removes inaccurate gender information from $\hat{V}_N$, as $\hat{V}_D$ contains limited semantic information other than gender. Semantic information from $\hat{V}_N$ becomes gender-neutral word embeddings. Their debiasing framework was based on the method proposed by [6], to remove confounding noise. It is called HalfSibling Regression (HSR). This required only one hyper-parameter and no word classification. $\hat{V}_N$ and $\hat{V}_D$ exchange gender information but not semantic information, hence the framework's name. $\hat{V}_N$ and $\hat{V}_D$ are half-siblings. HSR potentially functions effectively.

They tested their method on gender direction and gender-biased word relation tasks to determine the accuracy. They utilized gender direction relation tasks to try to figure out how much gender bias there is in terms of gender direction. Bias by Projection is a task determined by the dot product of the target word and gender direction. They provided the absolute values bias-by-projection of the embedding of the top 500 male and female biased keywords.

Moreover, they used SemBias which identifies the gender-definition word pair from four pairs, including a gender-biased word pair and two others. Gender-biased word relation tasks test word vector relation gender bias. They used GBWR Clustering which is a task that clusters the top 500 male-biased and female-biased words from the embedding into two clusters using k-means. GBWR-Correlation is also utilized to analyze the Pearson correlation amongst biases. GBWR-Profession calculates the 100 nearest neighbors of professional words proposed by Bolukbasi et al. [5] and Zhao et al [12]. GBWR-Association is also used similar to Gonen and Goldberg's use of the Word Embedding Link Test (WEAT) to design three associ-

ation tasks to explore the relationship between different female/male words. The GBWR-Classification task samples the top 5000 gender biased terms in this method. Their method also included one lexical-level task that tests gender-debiasing strategies.

Word similarity tasks compare word pair embeddings with human-annotated similarity levels. RG65, WordSim-353, Rarewords, MEN, MTurk-287, MTurk-771, SimLex-999, and SimVerb-3500 were the eight word similarity tasks they employed. Semantic textual similarity (STS) tasks evaluated sentence-level word embeddings through evaluating semantic comparability across two texts. Lastly, they included coreference resolution tasks to find all textual occurrences of a particular item. Two coreference resolution tasks—OntoNotes 5.0 and WinoBias have been tested in their approaches. Yang & Feng's [19] method does well on gender direction relation tests, gender-biased word relation tasks, lexical and sentence-level assessment tasks, and downstream coreference resolution tasks. In their method Hard-GloVe had the lowest bias-by-projection. But their word-vector-learning algorithm surpassed GN-GloVe. On SemBias, their post-processing approach performed best. HSR-GloVe stood second for GBWR-Clustering as it surpassed all post-processing and word-vector-learning approaches in terms of GBWR-Correlation, GBWR-Profession and GBWR-Association. HSR-GloVe had the lowest debiasing for GBWR-Classification. HSR-GloVe had five globally best coefficients in eight tasks, as it improved the eight-word similarity test coefficient by 6.96% and 7.31% in comparison to the initial embedding. HSR-GloVe performed best internationally on four of the five gender-debiasing tasks, and it outperformed the top postprocessing methods by 14.19%, well over 20 STS tasks, and the original embedding by 14.71%. In addition, HSR-GloVe had the lowest WinoBias-Diff between PRO and ANTI datasets as HSR-GloVe reduced WinoBias-Diff by 87.26% in contrast to Hard-GloVe.

Zhao et al. [20] explored how to detect and minimize gender bias in ELMo's contextualized word vectors in this research. Upon identification, the paper tried to fix the following issues: (1) The training data used by ELMo contained a disproportionate number of male entities compared to female entities, which resulted in a gender bias in the pre-trained contextual word embeddings; (2) the geometry of trained ELMo embeddings systematically encodes gender information; and (3) ELMo propagates gender information about male and female entities in unequal amounts. When properly studying how ELMo operates, it was discovered that the corpus used to train ELMo had a large gender bias. Male entities were over three times as prevalent than their female counterparts. As a consequence, the downloaded, pre-trained contextualized word embeddings had a gender bias. After further examination, it was shown that ELMo embeddings respond differently to male and female pronouns, with male entities being fourteen percent more accurately predicted from professional words than female entities were. The difference in accuracy between pro- and anti-stereotypical predictions was around thirty percent more in an ELMo-based system as compared to a GloVe-based system that is equivalent.

Using a training-time data augmentation strategy, they [20] added the gender-swapped variation of the corpus to the coreference system's training corpus (male entities are switched for female entities, and vice versa). The second strategy was a test-time embedding neutralization method that averaged input con-

textualized word representations with sentence word representations of the opposite gender. They first created a data model to apply the two strategies they investigated. The dataset had two unique subsets: pro-stereotype and anti-stereotype. In pro-stereotype, pronouns were connected to occupations that were mostly associated with the pronoun's gender. In the data augmentation approach, gender-revealing parts of the OntoNotes dataset were replaced with words indicating the opposite gender, and then the union of the original data and the swapped data were trained. In addition, they found it advantageous to decrease bias in supporting materials, therefore they replaced standard GloVe embeddings with bias-mitigated word embeddings. In contrast, the Neutralization technique constructed a gender-swapped version of the test cases as opposed to including gender-swapped words into the training corpus.

The original and gender-flipped phrases were then given contextualized word representations using ELMo, and the averages of these representations are utilized as the final representations. Finally, the findings indicated that data augmentation greatly minimizes the bias indicated by the coreference on WinoBias, but test time embedding neutralization is only somewhat effective. Although it required system retraining, data augmentation was mostly successful in removing bias from the ELMo-based coreference resolution method. Neutralization was only applicable to simpler circumstances since it is less successful than augmentation and cannot totally erase gender bias on the Semantics only portion of WinoBias.

Dev et al. [21] discussed measuring and mitigating biased inferences of word embeddings. They firstly measured the effect of specific biases. Secondly, they focused on bias attenuation on static word embeddings, using simple mechanisms. Lastly, they demonstrated that contextualized embeddings can be added to these approaches namely, ELMo [11] and BERT [15], but with limitations. To begin with, word embeddings have stereotyped meanings depending on the text they are trained on, which can enable downstream models that rely on them to reach incorrect conclusions.

Now for measuring bias, it relies mostly on the natural language inference (NLI) task. They built a vast number of examinations by populating a template with subject, verb, and object fillers in the following format: the subject, the verb, and an object or objects. They filled the verb and object spaces with a collection of activities that were performed often. Following that, the topic at hand is a collection of professions. This test has just one variable, and that is the topic. As a consequence of this, the final collection had one hundred eighty four objects, twenty seven verbs, a hundred and sixty four professions, and three gendered word pairs (man-woman, guy-girl, gentleman-lady). They expected the results to be neutral. Therefore, they defined three different measures for how far they are from neutral which are Net Neutral(NN), FractionNeutral(FN) and Threshold(T). About bias attenuation in static word embeddings, they focused on the simple projection operator [14] which simply identified a subspace associated with a concept hypothesized to carry bias, and then removed that subspace from all word representations. About bias attenuation in contextualized word embeddings, For ELMo, they attempted to attenuate biases by copying the previous projection method. In this procedure, they learned a bias subspace first and then removed it from the embedding. But

they only used it in layer 1. Additionally, they must execute this before the ELMo's BiLSTMs construct layers 2 and 3.

In BERT, they applied the same projection procedure they applied on ELMo. They concentrated on projecting the context-free subword BERT embeddings in a gendered direction to debias them. They were presented with two choices for when the debiasing should be executed. In the first option they adjusted the NLI model without debiasing and applied the constraint to debiasing during testing. On the other hand, in the second option they applied debiasing both during fine tuning the model and at test time.

From the results for measuring bias, it could be concluded that, out of the three embeddings used, BERT performed the best out of GloVe and ELMo. Even though BERT's performance was the highest, its results were far from the desired values. They used the results for bias attenuation in static word embeddings. By removing the derived subspaces from GloVe, they displayed a significant decrease in bias. Lastly, results for bias attenuation in contextualized word embeddings using ELMo showed that at layer 1, the ELMo embedding reduced intrinsic bias. Moreover, this improved predictor neutrality and lessened bias when used in gender-specific inference tasks. Moving on, results for bias attenuation in contextualized word embeddings using BERT demonstrated that the first option was completely ineffective at debiasing using NLI. However, the second option was much more effective at removing bias.

In this research, Sun et al. [18] discussed methods for reducing gender bias in NLP. They looked at recent research on gender bias in NLP detection and mitigation. To do this, they looked at methods for spotting bias and analyzed the phenomenon from the perspective of four distinct kinds of representational bias. There is evidence of gender bias in the training corpora, materials, pre-trained models, and algorithms designed for various models. When these components of NLP systems are biased, it may lead to erroneous predictions based on gender and, in some instances, even reinforce biases already present in the corpora the model is trained on [8].

They gave an introduction to gender bias assessment techniques and discussed the many kinds of representational biases that each technique detects. The Implicit Association Test (IAT) in psychology assesses people's unconscious gender bias. Based on this fundamental idea from the IAT, the Word Embedding Association Test was developed to quantify bias in word embeddings and it was found that even GloVe and Word2Vec embeddings contain human biases [7]. According to Bolukbasi et al. [5], the magnitude of an embedding's projection into the gender domain, which the embedding would employ to represent a gender-neutral phrase respresents gender bias and is correlated with the bias points given to the phrase by the impacted populations.

But according to Gonen and Goldberg [16], the technique falls short of capturing the full scope as words with comparable biases still tend to be represented by adjacent words in the embeddings. Gender bias is found in a model where if the model takes two sentences as inputs and gender-swapping is applied, there is a difference in the evaluation score. Retraining and inference are two categories of debiasing techniques they used. In contrast to retraining methods, which do not work without access to the original training data and require the model to be

retrained, inference methods can be used to remove bias at any time without the original training dataset.

They examined two families of methods for debiasing gender in word embeddings, one of which does not call for retraining and the other which does. i) Gender Subspace Removal: First, Schmidt [4] reduced gender-specific similarity by constructing a gender neutral framework utilizing cosine similarity and orthogonal vectors, although this might be troublesome as the meaning of a word could be intimately related to its gender portion. Later, Bolukbasi et al. [5] suggested changing the embedding space drastically by simply eliminating the gender element from gender-neutral words. ii) Pointing out the Gender-Neutral Words present in the Embeddings: Zhao et al. [12] invented a new way of debiasing embeddings and named it GN-GloVe. It makes a list of terms that are only used by one gender without a classifier. The writers develop the word embeddings by segregating gender-related information into a selection of dimensions and retaining information that is independent of gender in the remaining dimensions. They compiled current research on gender bias in NLP and how to recognize and reduce it in one publication.

For many applications, gender debiasing techniques in NLP are insufficient to completely debias models. They identified the following drawbacks of present strategies. First off, the majority of debiasing methods concentrate on a specific, modular NLP system operation. Second, it was unclear if the majority of gender debiasing techniques can be generalized to other tasks or models since their empirical validity has only been shown in a small number of applications [8]. Third, they pointed out that certain debiasing methods could impair performance by adding noise to an NLP model.

Gonen et al. [16] solved the question of the competence of existing bias removal techniques. Despite the fact that the results of previous debiasing methods may suggest that the bias has been significantly lowered according to the definition of bias provided in the papers, the actual impact is largely to conceal the bias. The gender bias is still reflected in the geometry of the "Gender-Neutral" words in the vector space. If existing bias removal techniques are insufficient, they cannot be trusted for developing gender-neutral modeling as the model will likely associate one implicit gendered term with another implicit gendered term. The gender bias of a word 'w' is described by its projection on the "gender direction," according to the standard definition provided by [5]. Training estimates the gender bias by averaging the disparities between female and male words in a fixed set, assuming that all vectors were normalized. The greater the scale of the projection, the higher the level of bias. To verify that all neutral terms were equally spaced from a pair of fundamentally gendered words, [5] employed a post-processing debiasing strategy that neutralized the gender projection of every term on a preset gender direction. [16] followed in the footsteps of [12], but adopted a different route. Rather than debiasing pre-existing word vectors, they modified the loss of the GloVe model [2] to concentrate majority of the gender characteristics within the last coordinate of a word. In this manner, the gender connotation of the term might be removed by dropping the last coordinate.

Even though gender-direction is a good indication of bias, it's not the sole factor that indicates bias. Even after using the above methods to remove bias, most

words that had a certain bias earlier were still grouped together, even though their gender direction has changed. This implied that the spatial geometry of the word embeddings remained mostly the same, except when it emerged to gendered words. Experiments that show that the methods for getting rid of bias don't work were also described. Using k-means to group the 1000 most biased words, they got the results for HARD-DEBIAS [5] and GN-GLOVE (gender-neutral GloVe) Clusters in the HARD DEBIASED embedding align with gender with 92.5% precision (based on the original bias of every word), compared to 99.9% precision in the initial biased version. The GN-GLOVE embedding achieves 85.6% accuracy, whereas the biased variant achieves 100%. These findings indicate that even after debiasing, most of the bias information remained ingrained in the representation. Clusters of words based on gender provide a novel approach to assessing bias. The fraction of target words that have gendered connotations among their nearest k neighbors. They could then compare the updated bias metric to the standard one and determine their relationship. The Pearson correlation for the HARD-DEBIASED embedding was 0.686. When neighbors were checked using the biased version, the correlation was 0.741. The Pearson correlation for the GN-GLOVE embedding was 0.736. (compared to 0.773). All of these associations had p-values below 0.05, signifying statistical significance. Using the neighbors-based bias definition, they created a graph representing a set of occupations. The amount of male neighbors represented on the Y axis, and the original bias was on the X axis. In HARD-DEBIASED, they obtain a Pearson correlation of 0.606 between the variables against a correlation of 0.747 when examining neighbors in accordance with the biased version and 0.792 (vs 0.820) in GN-GLOVE. The p-values for all of these associations are less than 1×10^{-30}.

After careful observation and analysis of the results of the experiments the writers came to the conclusion that the existing debiasing methods fail to completely remove bias from word embeddings. They found that words with strong previous gender bias clustered together. Moreover, they found that words that had a gender associated with it due to stereotypes clustered with other implicit gender words of similar gender. The inferred gender of words with previous gender stereotypes were easy to predict based on their vector geometry. Popular definitions used to quantify and eliminate bias were inadequate. Also, additional components of the bias in the vector geometry were taken into account and a complete debiasing method was needed to remove the bias in its entirety.

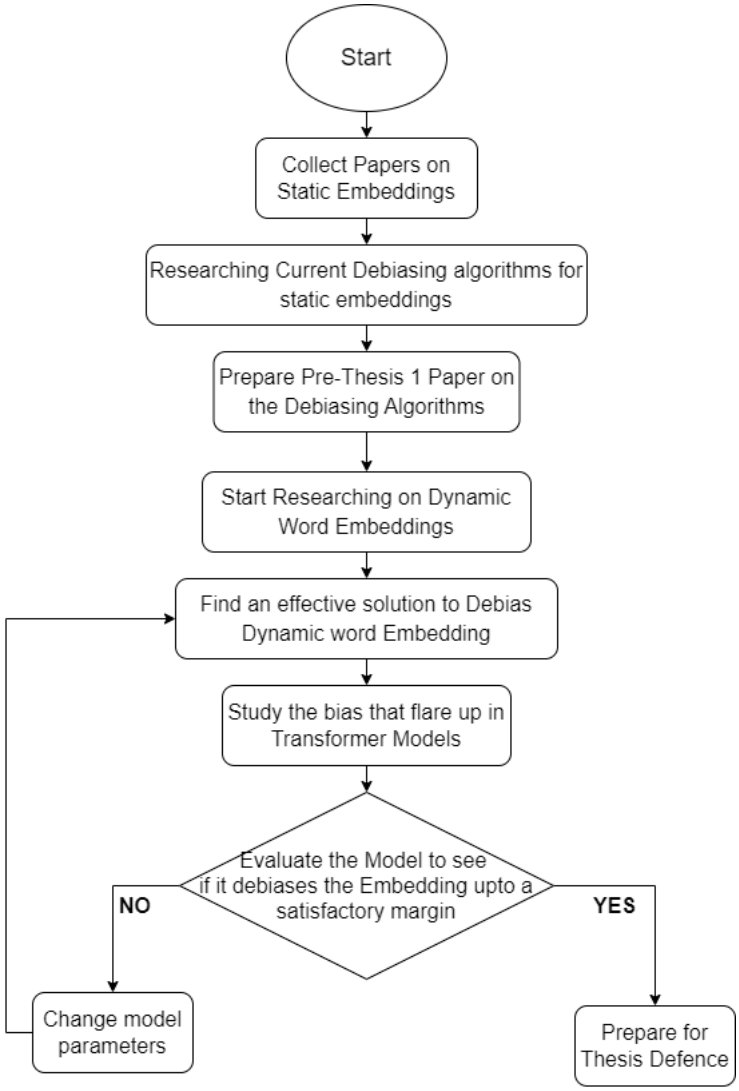
Chapter 4

Work Plan

For P1, we studied papers that focused on biases in word embeddings. We tried to understand how to capture biases that are implicit in pretrained static word embeddings. We learned about many debiasing algorithms: i) Hard Debias; ii) Double-Hard Debias; iii) Soft Debias; iv) Partial Projection; v) Flipping the Raw Test; vi) Linear Projection; vii) Subtraction; viii) RAN Debias.

In P2, we will shift our focus on dynamic word embeddings. There is currently very little work done on debiasing dynamic word embeddings. We also plan to work on the biases that flare up in many transformer models, like BERT.

In P3, we will continue fine tuning our debiasing algorithm for better results. We will evaluate the effectiveness of our debiasing algorithm through various tests like, analogy generation test, WEAT, embedding coherence and quality test etc.


Start
Collect Papers on Static Embeddings
Researching Current Debiasing algorithms for static embeddings
Prepare Pre-Thesis 1 Paper on the Debiasing Algorithms
Start Researching on Dynamic Word Embeddings
Find an effective solution to Debias Dynamic word Embedding
Study the bias that flare up in Transformer Models
Evaluate the Model to see if it debiases the Embedding upto a satisfactory margin
NO
YES
Change model parameters
Prepare for Thesis Defence

Chapter 5

Conclusion

In conclusion, it is possible to state that locating bias in word embeddings and developing strategies to combat it is a relatively new field of study in our day and age, but it is also a very essential one. In recent years, involvement in Natural Language Processing (NLP) has increased dramatically, and one might argue that word embeddings are the fundamental component of this field. Training data are used in the process of training a data model so that the model can reliably anticipate a variety of outputs. If there is bias in the learning data used to construct a system, obtaining the most accurate results over the long term becomes problematic. As a consequence, minimizing this bias from word embedding should be quite an intriguing topic to address. Because natural language processing (NLP) and, ultimately, word embeddings are used quite frequently in our everyday lives, and because artificial intelligence is becoming increasingly integrated in a variety of fields as a result of advances in technology, it should be our top priority to ensure that these systems are free of bias.

As a result of the literature review that we carried out, we came across many distinct methods of recognizing bias in word embeddings that were pointed out by various writers. It has been shown that a variety of biases, most notably those based on gender, are to blame for the skewed outcomes that are produced by the systems. Therefore, a number of studies pointed out how these biases may be reduced using the word embeddings that are offered in the pre-trained models, and as a consequence, the systems are able to produce outputs that are free of bias. Therefore, we believe that it is both vital and intriguing to find a solution to this issue, and we would want to offer our own ideas as well as a solution to the problem that we have discovered, which is identifying and mitigating gender bias from static and dynamic word embeddings using NLP.

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